K Kamal

ce24btech11030@iith.ac.in | ktkkamal06@gmail.com | +91 9966588918 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.85
Narayana Junior College	Class XII (Telangana state board of intermediate education)	2024	96.00%
Narayana school	Class X (Telangana state board of secondary education)	2022	10.00%

POSITION OF RESPONSIBILITY

Member of Inspection Team | Mess Monitoring Committee (MMC)

- **Grievance Redressal & Sentiment Mapping:** Overhauled the institutional dining framework by capturing structured user feedback across a cohort of 4,000+ students, driving down facility complaints by 40%.
- **Resource Optimization & Delivery Compliance:** Monitored vendor SLAs, hygiene standards, and operational food safety across two central dining facilities within a strict financial constraint of 130/day/student.
- **Policy & Menu Restructuring:** Translated qualitative dietary preferences into balanced operational guidelines, coordinating directly with administrative authorities and external catering contractors.

Head of Production & Operations | Milan General Championship Fest

- **Large-Scale Ground Execution:** Spearheaded crowd control, perimeter security, and end-to-end logistics for flagship pronites (Concerts, Prom Night, DJ) hosting 2,500+ active attendees.
- **Vendor Negotiations & Budget Management:** Led vendor procurement and contract negotiations for staging, audio-visual infrastructure, and food stalls, optimizing resource allocation while ensuring zero operational downtime.
- **Cross-Functional Team Leadership:** Managed multi-tiered volunteer and operations teams under high-pressure scenarios, mitigating live operational risks in real time.

Coordinator | Inter-IIT Sports Meet, IIT Hyderabad

- **Logistics & Stakeholder Management:** Directed end-to-end hospitality, travel, and onboarding for a delegation of **200+ visiting athletes**, acting as the primary point of contact for real-time grievance resolution and campus support.
- **Process Optimization:** Designed a structured resource allocation framework using Excel, Developed a structured allocation matrix to map visiting contingents by institution and sport.

Class Representative | Civil Engineering Batch, IIT Hyderabad

- **Stakeholder Liaison & Conflict Resolution:** Acted as the primary bridge between a **59-student cohort** and faculty administration; successfully negotiated complex timetable adjustments to resolve scheduling bottlenecks, specifically accommodating edge-case requirements (backlog students).
- **Process Tracking & Financial Auditing:** Centralized academic schedules using excel sheets and managed the verification and distribution of department-allocated financial reimbursements based on audited expense submissions.

PROJECTS

Event Operations & Financial Analytics Platform | C++, MySQL, Relational Database Modeling

- **Relational Database Design:** Architected a modular backend database with multi-table relational schema (foreign keys, junction tables) to manage vendors, venues, and human resources.
- **Financial Analytics & Reporting:** Automated core business intelligence metrics by deploying dynamic SQL aggregation queries to generate real-time dashboards tracking budgets, expense variances, and margins.
- **Data Integrity Safeguards:** Programmed robust validation logic and conflict detectors to prevent scheduling overlaps and resource misallocations.

Automated Data Pipeline & Predictive Modeling Framework | Python, Abaqus APIs

- **High-Throughput Batch Automation:** Built headless Python scripts to automate parameter mutations and batch-run 100+ multivariate simulation models, eliminating manual GUI dependencies and accelerating iteration cycles.
- **Algorithmic Boundary & Failure Testing:** Architected a Python-based binary search algorithm to automate iterative stress testing, rapidly pinpointing edge-case structural failure thresholds and dynamic load limits.
- **Large-Scale Data Ingestion & Optimization:** Engineered an automated data parsing pipeline to extract targeted metrics from multi-gigabyte binary databases, deploying dynamic cleanup routines to optimize processing throughput and disk I/O.
- **Predictive Analytics & Decision Frameworks:** Synthesized multi-variable dynamic impact data to generate predictive design envelopes, establishing empirical benchmarks and standardized engineering guidelines.

SKILLS & RELEVANT COURSES

Analytical & Data Tools: Python, SQL / Dynamic MySQL, Advanced Excel (Modeling & Auditing), Relational Data Architecture.

Governance & Strategy Competencies: Large-Scale Operations, Public Delivery & Quality Assurance, Stakeholder Management, Vendor Negotiation, Financial & Budget Auditing, Survey Feedback & Sentiment Analysis.

Problem Solving & Core Skills: Data Structures & Algorithms, Automated Data Pipelines, Parametric Modeling, Risk & Conflict Mitigation.

Languages: C, C++, SQL, HTML, CSS

Tools: VS Code, Git, GitHub, Canva

EXTRACURRICULAR

Volunteer, National Service Scheme: Facilitated large-scale public health and environmental student initiatives.

Events Executive, Civil Engineering Society: Orchestrated department-wide initiatives including Hackathons and the Tower Crane Competition.